

UC Cross-Atlas Reproducibility v1 Project Plan

A 12-month, two-author dry-lab project at UCSC

Contents

Cross-Atlas Reproducibility of GWAS-Driven Cell-Type Prioritization in Ulcerative Colitis	2
Part 1: The Question and Why It Matters	3
1.1 What you're studying	3
1.2 What this paper does <i>not</i> claim	3
1.3 Why ulcerative colitis specifically	4
1.4 Two-author division of labor	4
1.5 v2 trajectory and v1 architectural setup	4
1.6 Stretch goals (only added if locked core is clean)	6
1.7 What the deliverable looks like	6
1.8 The honest expected outcome	6
Part 2: Mini-Course	7
2.1 What a GWAS is, what summary statistics look like	7
2.2 From SNPs to gene scores: MAGMA	8
2.3 scRNA-seq atlas data: the AnnData object	9
2.4 The two methods	10
2.5 The reproducibility question, formally	13
2.6 Cell-type label harmonization	15
2.7 Brown's method for regime 2	16
2.8 Sanity scaffolding	17
2.9 Donor-level leave-one-out uncertainty intervals	18
2.10 Compute budget — locked v1 with stretches	18
2.11 Tooling stack	20
Part 3: Datasets	20
3.1 GWAS	20
3.2 scRNA-seq atlases	20
3.3 Reference data	21
Part 4: Month-by-Month Plan	21
M0 (pre-start): planning lock-in	21
M1: environment, GWAS, atlases	22
M2: cell-type harmonization	23
M3: regime 1 (both methods, both GWAS) + sanity scaffolding + fine-tier completion [Gate 1]	24
M4: concordance analysis (5 axes) + introduction draft	25

M5: regime 2 + broad-atlas comparators (HCA Gut + Pan-GI) + methods draft [Gate 2]	26
M6: write-up first draft + stretch decision	27
M7: top stretches if green-lit [Gate 3]	28
M8: lower-priority stretches + polish	28
M9: pre-submission peer review + revision	28
M10: journal selection + format	28
M11: submission	29
M12: respond to journal feedback	29
Three explicit fork points for descope decisions	29
Repo directory structure	29
Part 5: Risk Register	30
Part 6: Reading List	33
Part 7: How to Know You're On Track	34
DECISIONS.md template (commit at M0)	34
Final Note	38

Cross-Atlas Reproducibility of GWAS-Driven Cell-Type Prioritization in Ulcerative Colitis

v1: a 12-month, two-author dry-lab project at UCSC, with v2 architectural setup

Locked-core deliverable plus explicit stretch ladder. The locked core is a self-contained, publishable first paper. Stretches add depth only when the previous tier is clean — they are not fallbacks for a slipping core.

This version of the plan promotes four pieces from the original stretch ladder into the locked core (fine-tier granularity as primary, seismicGWAS as a second method, Liu 2023 multi-ancestry GWAS as cross-GWAS sensitivity, Pan-GI as a second broad-atlas comparator), **folds in three lightweight architectural decisions that enable v2 (UC + CD + RA cross-tissue generalization) without compromising v1 scope or schedule** (CL-ontology-aware harmonization at the immune-cell level in M2; seismicGWAS specificity score serialization in M3; donor-attribution metadata as a generalized pattern in M1), **pre-commits methodological choices on the operational specifications that would otherwise leak researcher-degrees-of-freedom mid-analysis** (Brown's correlation-matrix estimation, scDRS Z-score-vs-p-value policy, raw-counts policy for integrated comparators, bootstrap CIs on Spearman ρ , shared-cell-type intersection rule, scDRS covariates, all-cells policy, sub-clustering depth rule, marker-gene QC at end of M2, minimum cell-count threshold, detection threshold, seismicGWAS confounders, HCA donor-overlap verification, multiple-testing correction across the comparison battery, tied-rank handling), **and addresses immune-disease-specific and atlas-specific methodological details that a final round of scientific review surfaced** (MHC region exclusion from gene set, negative control GWAS, Mennillo pre-treatment-only subsetting, atlas preprocessing heterogeneity disclosure, Stouffer-style sample-size weighting acknowledgment for regime 2, headline random seed, λ_{GC} check, X chromosome exclusion, Cell Ontology version pinning, empirical correlation matrix edge case fallback).

Part 1: The Question and Why It Matters

1.1 What you’re studying

When researchers want to know which cell types in a tissue are genetically relevant to a disease, the standard approach combines two data sources: a genome-wide association study (GWAS), which tells you which variants associate with disease, and a single-cell RNA-seq atlas, which tells you which genes each cell type expresses. A class of computational methods — scDRS, scPagwas, seismicGWAS, MAGMA gene-property regression, FUMA, sc-linker — integrates these to nominate “disease-critical” cell types.

Recent work has explored this from several angles. Watanabe et al. 2019 (*Nat Commun*) used MAGMA gene-property regression on 43 scRNA-seq datasets across 26 traits, including IBD, but their corpus was heavily brain-biased (29 of 43 datasets) and predates modern UC colon atlases. Lakkis et al. 2024 (*Frontiers in Immunology*) ran scDRS, scPagwas, and scGWAS on UC, but used a single atlas. Li et al. 2025 (medRxiv 2025.05.24.25328275) benchmarked modern per-cell methods across 33 traits and 10 atlases, but each trait-atlas pair is run once — cross-atlas reliability for any single disease isn’t directly addressed.

What hasn’t been done: directly comparing the two operational regimes a practitioner actually chooses when applying GWAS-driven cell-type prioritization to a disease with multiple available atlases, with cross-method validation, cross-GWAS validation, and at the granularity where disease-relevant biology lives.

- **Regime 1 (single-atlas):** Run scDRS or seismicGWAS on whichever atlas you have.
- **Regime 2 (per-atlas-then-meta-analyze):** Run each method separately on each atlas, then combine cell-type p-values across atlases via correlation-aware meta-analysis (Brown’s method, which extends Fisher’s method to non-independent inputs).

We compare these regimes side-by-side on three independent UC colon atlases (Smillie 2019, Kong 2023, Mennillo 2024) against two UC GWAS (de Lange 2017, EUR; Liu 2023, multi-ancestry), using two methods (scDRS, seismicGWAS) at two granularity tiers (broad ~10–15 cell types; fine ~30–50 cell states). Within regime 1, we additionally quantify cross-atlas concordance via Spearman ρ , top-k Jaccard, and Cohen’s κ on FDR-significance, plus cross-GWAS within-atlas concordance and cross-method within-atlas concordance.

1.2 What this paper does *not* claim

- *Not “regimes generalize across diseases.”* This is a UC-only paper. Cross-disease and cross-tissue generalization are v2’s job.
- *Not “regime 3 (integration) is gold standard.”* Regime 3 isn’t in the locked v1 at all. Deferred to v2.

- *Not “concordance measures whether the same biology is identified.”* The scDRS group test uses an atlas-specific reference distribution; some apparent cross-atlas disagreement is structural, not biological.
- *Not “scDRS and seismicGWAS are statistically independent methods.”* Both consume the same upstream MAGMA gene-level Z-scores. Cross-method agreement is partly mechanical; cross-method disagreement is more informative.
- *Not “Pan-GI is an independent broad-atlas replication of our UC trio.”* Pan-GI integrated Smillie 2019 and anchored Kong 2023, so 2/3 of our core atlases’ donors are inside Pan-GI. We frame Pan-GI specifically as a robustness-against-integration-pipeline-choices comparator. **HCA Gut is the only fully-independent broad comparator.**
- *Not “Liu 2023 cross-GWAS robustness implies the regimes generalize across diseases or ancestries.”* Liu 2023 UC is the same disease in a partially overlapping disease cohort. It’s a cross-GWAS sensitivity within UC, not a cross-trait test.

1.3 Why ulcerative colitis specifically

Three reasons. The IBD/UC GWAS landscape is large and stable — de Lange 2017 has full summary statistics publicly available, Liu 2023 added a multi-ancestry meta-analysis with even larger N, and the underlying genetics isn’t in flux. The colon mucosa has at least three independent, high-quality, public scRNA-seq atlases generated by different labs with somewhat different protocols. UC has been used as a test case in Lakkis 2024, so this work positions naturally as the cross-atlas extension.

UC is also the right starting point for the longer-term trajectory toward a UC + CD + RA cross-tissue generalization paper. The infrastructure built here — pipeline, harmonization vocabulary, figures, manuscript scaffolding — carries forward to v2.

1.4 Two-author division of labor

This is a two-author project. Document the division explicitly in the README before any code is written. Some pieces are unambiguously joint (cell-type harmonization, regime 2 implementation, the manuscript). Method-axis split is now natural with two methods in the locked core. Authorship: whoever drafts the manuscript is first author; co-first if contributions are genuinely indistinguishable. Discuss with PI mentor before work starts. Revisit the division at month 4.

1.5 v2 trajectory and v1 architectural setup

v2 is **not** UC + CD alone — it is **UC + CD + RA cross-tissue generalization**, an 18–24 month follow-up project that uses regime 3 (scANVI integration) as its primary methodological contribution. The headline claim v2 is positioned to make:

“Within-disease cross-atlas reproducibility extends to cross-disease comparison (UC vs CD) at shared immune cell lineages but fails for tissue-specific stromal populations. Cross-tissue generalization (gut vs synovium) requires multi-atlas integration (regime 3) and remains incomplete for tissue-specific cell types.”

That’s a substantive, falsifiable claim that justifies a separate paper. Within-IBD (UC + CD alone) was too genetically similar to support a real generalization test; adding RA puts v2 at a different disease class, different tissue, different cell types altogether.

Three v1 architectural decisions enable v2 cleanly. They cost essentially nothing in v1 if planned in; they cost weeks if retrofitted later. All three go in DECISIONS.md at M0.

1. Cell-type harmonization with cross-tissue extension in mind (M2). When building the gut mucosa vocabulary, immune cell types are mapped to **Cell Ontology terms whose parent terms align with synovium-applicable categories**. Concretely: the gut “T cell” and a hypothetical synovium “T cell” should share parent CL terms (e.g., CL:0000084 “T cell” → CL:0000542 “lymphocyte”). When in doubt during M2 manual mapping, prefer the higher CL term that admits both a gut and a synovium daughter. Cost: ~1 day of CL ontology browsing during M2 harmonization.

2. seismicGWAS specificity score serialization (M3). seismicGWAS specificity scores are GWAS-independent (they describe the atlas, not the disease) and are the substrate for v2’s cross-tissue cell-type comparison (“how does UC-T-cell specificity compare to RA-T-cell specificity for the same gene set?”). Save specificity scores per (atlas, granularity, cell_type, gene) as serialized R objects (.rds) or feather files, **not** embedded in the regression-result objects. Schema: a long-format dataframe with columns atlas, cell_type, granularity, gene, specificity_score. Saved alongside regression results. Cost: ~half a day of pipeline organization during M3 seismicGWAS work.

3. Donor-attribution metadata as a generalized pattern (M1). v1 already requires per-donor source attribution for Pan-GI to support the donor-overlap-exclusion analysis. Build this as a **generalized pattern**, not a Pan-GI special case: for every atlas, save a metadata table mapping every donor ID to its originating study (Smillie 2019, Kong 2023, Mennillo 2024, Pan-GI internal, AMP-RA Phase 1, etc.). Format: donor_id, originating_study, tissue, disease_status, assay_protocol. This generalizes to AMP-RA Phase 2 in v2, which integrates several earlier RA cohorts. Cost: minimal beyond the Pan-GI work already required.

What v1 explicitly does *not* do for v2: - *Does not run any RA atlas*. RA atlases enter at v2 only. - *Does not run regime 3*. scANVI integration is v2’s central methodological contribution. - *Does not commit to specific RA atlases or RA GWAS*. That decision happens at v2 planning time, when you’ll know which atlases are accessible and what the v1 results suggest about likely v2 challenges. - *Does not pay any v1 cost beyond the three architectural decisions above*. If the PI does not commit to v2 trajectory, drop the three decisions; v1 stands alone fine.

The PI commitment question. v2-combined is an 18–24 month project after v1 ships, totaling a 30–36 month research line. The three v1 architectural decisions are cheap *if* v2 is going to happen. They are wasted effort if v1 is the only output. Get explicit PI sign-off on the v2 trajectory before committing the v1 setup work.

1.6 Stretch goals (only added if locked core is clean)

Stretch ladder is compact — most of the original stretches were absorbed into core. Stretch N+1 only starts when stretch N is fully clean and written up.

1. **Brown’s-method permutation-based covariance.** Conditional fallback: activate if the empirical Kost-McDermott correlation matrix from M3 null draws shows pathological entries (negative or near-zero off-diagonals where positive is expected). With empirically-estimated correlations replacing the original under-specified analytical procedure, this fallback is much less likely to be needed. ~300 additional scDRS runs if activated.
2. **scANVI integration as regime 3.** Deferred to v2 by default — only attempt if locked core is done by M5 *and* PI explicitly trades v2 differentiation for v1 strength. Generally: don’t.
3. **LDL × Tabula Sapiens hepatocytes complementary positive control.** Closes the human-only-code-path gap. ~1 day.

scPagwas is not in v1 at all (compute infeasible). The de Lange IBD-combined sensitivity is not in v1 (Liu 2023 is the actually-strong cross-GWAS sensitivity).

1.7 What the deliverable looks like

A bioRxiv preprint targeting *Bioinformatics Advances* (primary) or *NAR Genomics and Bioinformatics* (secondary). Both accept methods-comparison papers, undergrad first/co-first authors, and 2–4 month review cycles. *BMC Bioinformatics* is a viable third option.

The narrative: “We compared single-atlas (regime 1) and meta-analyzed (regime 2) operational regimes for GWAS-driven cell-type prioritization on three independent UC colon atlases, evaluated against two UC GWAS (de Lange 2017 EUR; Liu 2023 multi-ancestry), using two methods (scDRS, seismicGWAS) at two cell-type granularity tiers. Within regime 1, [X]% of broad cell types and [Y]% of fine cell states replicated across atlases. Cross-GWAS within-atlas concordance was [high/moderate]. Regime 2 [stabilized / changed / preserved] regime 1 conclusions. Cross-method concordance within atlas [exceeded / matched / fell below] cross-atlas concordance within method. We additionally compared to two broad multi-tissue gut atlases — HCA Gut Cell Atlas (independent of our core trio) and Pan-GI (which integrates Smillie 2019 and anchors Kong 2023, providing an integration-pipeline-robustness comparison) — and ran a parallel MAGMA gene-property sanity track.”

Realistic publication timeline: submit M11–M12 of this plan; first decision M14–M15; published M17–M19.

1.8 The honest expected outcome

Modal result for highly polygenic immune diseases like UC is broad-tier saturation: most cell types pass FDR < 0.05 in every atlas. Promoting fine tier into the locked core specifically addresses this — fine tier is much less likely to saturate.

Likely outcomes across the new dimensions:

- *Cross-atlas (within-method, within-GWAS)*: broad tier likely saturates; fine tier likely shows some replicating cell states (immune cells, mature enterocytes) and some atlas-specific cell states.
- *Cross-method (within-atlas, within-GWAS)*: broad tier likely high concordance (shared MAGMA inputs); fine tier may show divergence.
- *Cross-GWAS (within-atlas, within-method)*: likely high concordance — de Lange and Liu are both UC GWAS. The interesting case is cell types where they disagree.
- *HCA Gut (broad comparator, independent)*: likely lower concordance with UC trio than within-trio concordance, especially at fine tier.
- *Pan-GI (broad comparator, donor-overlapping)*: with donor overlap, very high concordance with Smillie + Kong (mostly the same data); with donor-overlap exclusion, drops toward HCA-Gut-like levels — the informative comparison.

The contribution is incremental and defensible: cross-atlas concordance numbers don't currently exist for UC at this granularity, the regime-1-vs-regime-2 comparison hasn't been published for UC, cross-GWAS sensitivity hasn't been quantified, and integration-pipeline-robustness via Pan-GI is novel as a framing.

Part 2: Mini-Course

2.1 What a GWAS is, what summary statistics look like

A genome-wide association study tests, for each of ~8-12 million common SNPs, whether one allele vs. the other associates with the trait. For each SNP: effect size, standard error, p-value. Modern complex-trait GWAS routinely identify hundreds of genome-wide-significant hits at $p < 5 \times 10^{-8}$.

Most signal is sub-significant but real. UC is highly polygenic. Cell-type prioritization methods exploit aggregate signal across many genes — they need full summary statistics.

SNPs are correlated. Linkage disequilibrium (LD) is the central statistical headache of post-GWAS analysis.

Two GWAS in locked v1:

- **de Lange et al. 2017 UC** — primary. GWAS Catalog GCST004131. ~12,366 cases / ~33,609 controls, EUR ancestry.
- **Liu et al. 2023** — cross-GWAS sensitivity. *Nat Genet* 55:796-806. Multi-ancestry IBD meta-analysis (East Asian + European). UC arm only.

Quality-check three things on each download: genome build (should be **GRCh37 / hg19**), allele consistency (A1/A2 clearly labeled), imputation quality (drop INFO < 0.6 and MAF < 0.01).

Sanity check: top UC genes should include IL23R, JAK2, TYK2, NKX2-3, IL12B, SMAD3, STAT3, MHC region. NOD2 should be weaker than for CD. Liu 2023 should produce broadly similar top genes — if top genes are unrelated to immunity, you've grabbed the CD or IBD-combined arm by mistake.

2.2 From SNPs to gene scores: MAGMA

Use **MAGMA** (de Leeuw 2015), precompiled C++ binary from CTGLab.

Step 1 — SNP-to-gene annotation. 10 kb upstream + 10 kb downstream window, standard.

```
magma --annotate window=10,10 --snp-loc snps.loc --gene-loc NCBI37.3.gene.loc --out ste
```

Step 2 — Gene-based test. LD reference must match GWAS ancestry.

- de Lange (EUR): 1000 Genomes Phase 3 EUR.
- Liu 2023 (multi-ancestry): EUR is a clean first-pass approximation. The alternative (per-ancestry MAGMA + meta-combine) is a research project. Document choice in DECISIONS.md and acknowledge approximation in Methods.

```
magma --bfile g1000_eur --pval gwas_uc_delange.pval N=45975 --gene-annot step1_annot.ge  
magma --bfile g1000_eur --pval gwas_uc_liu.pval N=NNNNN --gene-annot step1_annot.genes.
```

Step 3 — Convert to scDRS gene set. Two .gs files (one per GWAS) plus two gene-level Z-score tables (seismicGWAS uses Z-scores directly).

MHC region exclusion (critical for immune disease GWAS). UC has the strongest GWAS signals in the MHC region (chr 6, ~28.5–33.5 Mb in GRCh37 — HLA-DRB1, HLA-DQA1, HLA-DPB1). MHC’s complex LD structure means a single causal variant can produce ~50+ correlated genome-wide-significant SNPs, and multiple genes within the MHC inherit overlapping signal. If MHC genes enter the top-1000 gene set unfiltered, scDRS scores can be dominated by MHC-driven cells (mostly antigen-presenting cells), confounding cross-atlas comparison.

Pre-commit: exclude MHC genes from both the scDRS top-1000 gene set and the seismicGWAS gene-Z-score table. Define MHC as chr 6: 28,477,797–33,448,354 (GRCh37 coordinates per the IMGT/HLA region definition). Identify MHC genes via the gene annotation file used by MAGMA (NCBI37.3.gene.loc). Apply the filter *after* MAGMA gene-based test but *before* .gs file generation, so MAGMA still sees MHC SNPs but downstream scoring excludes MHC genes.

Sensitivity analysis with MHC retained. Run a single supplementary scDRS analysis on Smillie under de Lange with MHC genes included; report cell-type rankings vs the MHC-excluded headline. Confirm MHC inclusion shifts antigen-presenting cell rankings as expected. Locked in DECISIONS.md.

X chromosome handling. Restrict MAGMA gene-based test to autosomes only (chromosomes 1–22). Standard practice in cell-type prioritization; X-chromosome handling in MAGMA requires sex-stratified analysis that’s not present in our GWAS summary statistics. Documented in Methods. Locked in DECISIONS.md.

Liu 2023 sample-size N column. Multi-ancestry GWAS often have per-SNP N varying across SNPs because different ancestries contribute different variants. Verify in M1 whether Liu 2023 summary statistics include a per-SNP N column. If yes, use the column. If no, document the fixed-N approximation in DECISIONS.md and Methods.

λ_{GC} sanity check. After MAGMA SNP-to-gene annotation, compute genomic in-

flation factor λ_{GC} for both GWAS. Report in DECISIONS.md. If $\lambda > 1.1$, consider applying LDSC-corrected MAGMA in revision (not as a pre-submission requirement; flag for reviewer response).

2.3 scRNA-seq atlas data: the AnnData object

AnnData object

— X	cell × gene count matrix (sparse)
— obs	cell metadata: cell_type, donor_id, disease_status, QC
— var	gene metadata
— layers	alternative count matrices

Counts vs. normalized. scDRS expects raw counts in `adata.X`. seismicGWAS works with log-normalized in a `SingleCellExperiment` with `logcounts`.

Critical: gene symbol drift. Atlases from different years use different HGNC releases. Pin a single release (e.g., HGNC 2024-Q1) and remap every atlas. Verify ~95%+ of canonical UC GWAS hits (IL23R, JAK2, TYK2, NKX2-3, ATG16L1) appear under the same symbol after harmonization.

Atlas preprocessing heterogeneity policy. Smillie 2019, Kong 2023, and Mennillo 2024 were preprocessed by their original authors with different doublet-removal protocols (Scrublet vs DoubletFinder vs none), different ambient-RNA correction (SoupX vs CellBender vs none), different cell-cycle handling, and different QC thresholds.

Pre-commit Option B (pragmatic): take published preprocessing as-is, do not re-run uniform doublet/ambient/QC across atlases. The marker-gene QC step at end of M2 (§2.6) catches the worst harmonization problems caused by upstream preprocessing differences. Disclose in Methods Limitations: “Cross-atlas concordance estimates are partly confounded by published-atlas preprocessing differences (doublet removal, ambient RNA correction, QC thresholds). The marker-gene QC step at end of M2 mitigates the worst label-mismatch problems caused by these differences but does not eliminate them.” Locked in DECISIONS.md.

Mennillo timepoint policy. Mennillo 2024 is an anti-integrin therapy time-course atlas with samples taken before and after treatment. Pre-commit: **use Mennillo pre-treatment baseline samples only** for the locked v1 analysis (most comparable to Smillie/Kong active-UC samples). Document loss in donor count after subsetting. **If pre-treatment N is below 8 donors, reconsider Mennillo’s role at M1 and consider replacing with Garrido-Trigo 2023 from the fallback list.** Locked in DECISIONS.md.

Sample composition disclosure (inflamed vs uninflamed). Smillie 2019 includes both inflamed and non-inflamed regional samples; Kong 2023 may include both; Mennillo (after pre-treatment subsetting) may include both. The all-cells policy (§2.4.1) handles this — no further subsetting — but the inflamed/uninflamed composition per atlas must be documented in Methods Table 1 (donor counts, sample counts by inflamed/uninflamed status, per atlas). Locked in DECISIONS.md.

2.4 The two methods

2.4.1 scDRS (Zhang et al. 2022, *Nature Genetics*) For each cell c and trait gene set G :

$$\text{raw_score}(c) = \sum_g [w_g \times \text{normalized_expression}(c, g)]$$
$$\text{score}(c) = (\text{raw_score} - \text{mean of 1000 control gene set scores}) / \text{sd of control scores}$$

The matched-null (1,000 control gene sets matched on mean expression and variance) is the key statistical innovation.

Group analysis uses the full atlas as reference distribution — caveat in §2.4.3.

```
# Once per (atlas, GWAS) – generates per-cell scores
scdrs compute-score --h5ad-file data/atlasses/smillie.h5ad \
  --gs-file data/gwas/uc_delange_top1000.gs \
  --cov-file data/atlasses/smillie_covariates.tsv \
  --out-folder results/scdrs/smillie_delange/

# Group analysis at broad (post-processing, fast)
scdrs perform-downstream --score-file results/scdrs/smillie_delange/UC.score.gz \
  --group-analysis cell_type_broad --out-folder results/scdrs/smillie_delange_broad/

# Group analysis at fine (same per-cell scores)
scdrs perform-downstream --score-file results/scdrs/smillie_delange/UC.score.gz \
  --group-analysis cell_type_fine --out-folder results/scdrs/smillie_delange_fine/
```

compute-score is the heavy step (~30–45 min, 15–20 GB RAM). Per-cell scores feed both granularity outputs. **Fine-tier promotion adds essentially zero compute at the scoring level.**

For cross-GWAS sensitivity: 2 compute-score runs per atlas (de Lange, Liu), 4 group-analysis outputs.

Cross-atlas headline metric uses Z-scores, not p-values. scDRS p-values are computed against an atlas-specific Monte Carlo null distribution; same biology in two atlases produces different p-values purely from compositional differences. Z-scores are the pre-Monte-Carlo statistic and are far more comparable across atlases. Headline cross-atlas Spearman ρ uses cell-type-level Z-scores (mean per-cell Z within cell type). Within-atlas FDR-significance counts and Cohen’s κ stay on p-values, where the within-atlas Monte Carlo null is appropriate. This is a methodological commitment locked in DECISIONS.md.

Save the per-null-draw cell-type Z-scores. scDRS internally generates ~1000 null gene-set Monte Carlo draws and computes per-cell null scores. We aggregate these to cell-type-level null statistics (mean Z within cell type, per null draw, per atlas) and serialize them. These are the inputs to Brown’s method correlation-matrix estimation in M5 (see §2.7). Format: feather file per (atlas, GWAS, tier) with rows indexed by (null_draw_idx, cell_type). Cost: post-processing on existing scDRS outputs.

Raw counts policy for Pan-GI and HCA Gut. Both are integrated atlases with cell-type labels assigned post-integration. We use the **raw count layer** (e.g.,

`adata.layers['counts']` from cellxgene Discover, or `adata.raw.X` if available) with **published cell-type labels** — no re-clustering. Methods text explicitly limits Pan-GI/HCA results to “broad-comparator robustness check,” not independent validation. Verify raw counts are accessible during M1 atlas standardization; if not, drop the comparator to stretch.

Covariate set in compute-score. scDRS regresses out cell-level covariates before the Monte Carlo scoring. The choice of covariates affects which cell types light up — particularly for cell types whose abundance correlates with QC metrics (plasma cells often have unusual gene/count profiles, for example). We pre-commit the following covariate set, applied consistently across all 5 atlases:

- `log_n_genes` — log10 of detected genes per cell
- `log_n_counts` — log10 of total counts per cell
- `donor` — one-hot encoding of donor ID
- `sample` — one-hot encoding of sample ID, included only if the atlas has multiple samples per donor
- `sex` — one-hot encoding (M / F / unknown), included if donor-level sex metadata exists in the atlas. UC has known sex-biased prevalence and immune-cell composition differs by sex; not regressing it out can confound cell-type rankings. If sex metadata is missing for an atlas, document the omission in DECISIONS.md and Methods.

Generated per atlas in M1 as `<atlas>_covariates.tsv`. Locked in DECISIONS.md.

All-cells policy. scDRS compute-score and group analysis run on **all cells in the atlas, regardless of disease status**. Smillie includes both UC and healthy donors; Kong includes both; Mennillo includes UC at varying treatment stages. Running on the full mucosa as reference is consistent with Zhang et al. 2022’s standard practice and is the right answer to the biological question (“which cell types in colon mucosa carry UC genetic signal”). Subsetting to UC-active cells only would change the Monte Carlo null distribution and shift cell-type rankings in ways that confound the cross-atlas comparison. Locked in DECISIONS.md.

Minimum cell-count threshold per cell type. Cell types with fewer than 50 cells in any atlas in a given comparison are excluded from concordance metrics for that comparison. This applies to all five concordance axes. Cell types below threshold are reported in supplementary tables with cell counts but do not enter the headline figures. Locked in DECISIONS.md.

Random seed for the headline analysis. Test-retest uses 3 seeds (1, 2, 3); the **headline scDRS run uses seed=42** for the Monte Carlo null sampling. Bootstrap CIs use seed=42 for the resampling. All random-seed-dependent steps are reproducible by re-running with this seed. Locked in DECISIONS.md.

2.4.2 seismicGWAS (Lai et al. 2025, *Nature Communications*) Specificity score for gene g in cell type c : avg expression of g in cells of type c , normalized by gene’s mean expression across all cells, with a variance-stabilizing transform. Within-atlas robustness; cross-atlas robustness is your project’s question.

For each cell type:

$$Z_g \sim \beta_0 + \beta \times \text{specificity}(g, c) + \varepsilon$$

Cross-GWAS is conceptually clean: specificity is computed *once per atlas* (GWAS-independent), regressed against de Lange Z-scores and Liu Z-scores separately.

```
library(seismicGWAS)
library(SingleCellExperiment)

sce <- readRDS("data/atlasses/smillie_sce.rds")

# Once per atlas – GWAS-independent
spec_broad <- calc_specificity(sce, ct_label_col = "cell_type_broad")
spec_fine  <- calc_specificity(sce, ct_label_col = "cell_type_fine")

# Once per (atlas, GWAS, granularity)
res_broad_delange <- get_ct_trait_associations(spec_broad, magma_z_delange)
res_fine_delange  <- get_ct_trait_associations(spec_fine,  magma_z_delange)
res_broad_liu     <- get_ct_trait_associations(spec_broad, magma_z_liu)
res_fine_liu      <- get_ct_trait_associations(spec_fine,  magma_z_liu)
```

Specificity score serialization (v2 setup decision). Save specificity scores as a long-format .rds (or feather) file separately from regression results. Schema:

```
# specificity_long.rds – one such file per atlas
data.frame(
  atlas      = "smillie",
  granularity = c("broad", "fine"),
  cell_type  = "T_cell",
  gene       = "IL23R",
  specificity = 0.42
)
```

Required because v2’s cross-tissue claim needs to compare UC-T-cell specificity to RA-T-cell specificity for the same genes. Embedding specificity inside regression-result objects forces re-running everything later.

Confounders in get_ct_trait_associations(). seismicGWAS regresses gene-level GWAS Z-scores against cell-type specificity. Without confounders, results are biased — long genes accumulate more GWAS signal because they contain more SNPs, and high-LD genes do similarly. We use the package’s default confounder set: **gene length (log-transformed), gene-gene LD score, and transcript count.** Verify in M1 by inspecting the seismicGWAS package source (the defaults to get_ct_trait_associations()) and confirm these are applied. Document the confounder set in DECISIONS.md and the Methods section. If the package defaults differ from this set, override and apply this explicit set.

Permuted-Z-score null draws for Brown’s method. Run $M = 1000$ permutations of the gene-Z-score vector, re-run get_ct_trait_associations() under each permutation, save the per-permutation cell-type-level test statistics to a feather file with rows indexed by (permutation_idx, cell_type). Used in M5 for Brown’s method correlation-matrix estimation (see §2.7). Cost: ~30 min per atlas-GWAS-tier; ~5

hours total for the locked core.

2.4.3 Structural caveats Methods are not statistically independent. Both consume the same upstream MAGMA gene-level Z-scores. Cross-method agreement is partly mechanical; cross-method disagreement is more informative. State explicitly in Methods.

scDRS group analysis uses an atlas-specific reference distribution. Two atlases with different cell-type compositions can give different group-level p-values for biologically identical populations.

scDRS requires raw counts; seismicGWAS works with log-normalized.

Disease-vs-healthy tissue caveat. UC-affected tissue contains inflammation-induced cell states (BEST4+ disease-state enterocytes, inflammatory fibroblasts) that healthy reference atlases (HCA Gut) lack.

Cross-atlas concordance is computed on the shared cell-type intersection, with a minimum cell-count threshold. When comparing two atlases that don't share all cell types — e.g., Smillie (UC + healthy, has inflammatory fibroblasts) vs HCA Gut (healthy only, no inflammatory fibroblasts) — we compute Spearman ρ , Jaccard, and κ on the **intersection of cell types harmonized to common labels** in both atlases. Each cell type in the intersection must have **≥ 50 cells in both atlases** to enter the concordance calculation. Atlas-specific cell types are reported separately as “detected in atlas X but absent from atlas Y” — they do not enter the concordance calculation, in either direction. This rule applies to all five concordance axes and is locked in DECISIONS.md.

“Detected” is operationally defined. A cell type is “detected” in an atlas when (a) its harmonized label is present in the atlas's crosswalk AND (b) the atlas has ≥ 50 cells assigned to that label. “Absent” means at least one of these conditions fails. Reporting distinguishes “label absent from crosswalk” (truly not labeled in the atlas) from “label present but below cell-count threshold” (small population, may exist biologically but unreliable for scDRS group analysis).

Pan-GI donor-overlap caveat. Pan-GI integrated Smillie 2019 and anchored Kong 2023 — 2/3 of core atlases' donors are inside Pan-GI. Frame as integration-pipeline-robustness comparator only, with and without donor-overlap exclusion. Important framing precision: **donor-overlap-excluded Pan-GI is still an scVI-integrated multi-study atlas** — just without our trio's donors. It is not “Pan-GI without integration.” It tests integration-pipeline robustness against varied study composition, not unintegrated cross-cohort robustness. Methods text states this explicitly. **HCA Gut is the only fully-independent broad comparator** — pending M1 verification that HCA Gut's donor set has zero overlap with Smillie/Kong/Mennillo (see §M1 spot-checks).

2.5 The reproducibility question, formally

Five concordance axes:

1. **Cross-atlas (within method, GWAS, tier).** 3 atlas pairs $\times$ 2 methods $\times$ 2 tiers $\times$ 2 GWAS = 24 ranking comparisons.
2. **Cross-method (within atlas, GWAS, tier).** 3 atlases $\times$ 2 GWAS $\times$ 2 tiers = 12 comparisons.
3. **Cross-GWAS (within atlas, method, tier).** 3 atlases $\times$ 2 methods $\times$ 2 tiers = 12 comparisons.
4. **Regime 1 vs regime 2 (within method, GWAS, tier).** 2 methods $\times$ 2 tiers = 4 comparisons (regime 2 on de Lange only).
5. **Tissue-matched vs broad atlas, two flavors.** HCA Gut (independent, pending M1 donor-overlap verification) and Pan-GI (re-integration with known donor overlap, dual analysis with/without exclusion).

Three metrics, with explicit handling rules:

(1) Headline metric: Spearman ρ on cell-type-level test statistics, NOT p-values. For scDRS: mean per-cell Z-score within cell type. For seismicGWAS: regression coefficient. Z-scores and coefficients are atlas-comparable; p-values are not (see §2.4.3 for the within-atlas-null caveat). Computed via `scipy.stats.spearmanr` (Python) or `cor(method="spearman")` (R), both of which use **average-rank tie-breaking by default** — locked in DECISIONS.md. Reported on the **shared cell-type intersection** between the two atlases (see §2.4.3).

(2) Top-k Jaccard. $k = 5$ and $k = 10$ at broad tier; $k = 5, 10, 20$ at fine tier. Top-ranked cell types by Z-score (scDRS) or regression coefficient (seismicGWAS). On shared intersection.

(3) Cohen's κ on FDR-significance. Within-atlas Benjamini-Hochberg FDR; κ on FDR-significance call (significant vs not). Marginal-saturation contingency: report κ at FDR < 0.01 as headline if $\geq 80\%$ saturation at FDR < 0.05 . On shared intersection.

Bootstrap 95% confidence intervals on every reported Spearman ρ . 1000 iterations, resampling over cell types within the shared intersection. **Percentile method** (2.5th and 97.5th percentiles of the bootstrap distribution); BCa (bias-corrected accelerated) deferred to revision if reviewers request. Bootstrap uses seed=42. Cheap (~seconds per comparison). Reported alongside the point estimate in every figure and table. Locked in DECISIONS.md.

Multiple-testing correction across the comparison battery. The single primary analysis is the 3×3 atlas-pair Spearman ρ heatmap at broad tier under de Lange via scDRS — reported uncorrected. **All other comparisons (the remaining ~60 across the four other concordance axes) receive Benjamini-Hochberg FDR correction at FDR < 0.05 across the full battery for any claim of statistical significance.** Uncorrected p-values reported in supplementary tables for transparency. Locked in DECISIONS.md.

Statistical-power note. Broad-tier comparisons ($n \approx 10$ –13 cell types after intersection) have wide Spearman ρ confidence intervals — at $n=10$, the SE on ρ is roughly 0.3, meaning $\rho=0.6$ vs $\rho=0.3$ may not be statistically distinguishable. Bootstrap CIs are the headline uncertainty quantification for sampling variance over cell types; donor-LOO is a separate axis (variance from donor sampling within atlas). Fine-tier comparisons ($n \approx 25$ –40) carry the discriminating power for between-axis claims. Discuss

explicitly in Methods and in Limitations.

The headline 3×3 heatmap is summarizing three pairwise ρ values. The lower triangle of the 3×3 atlas-pair matrix has only three entries: Smillie-Kong, Smillie-Mennillo, Kong-Mennillo. Pattern claims across these three values (e.g., “Smillie-Kong concordance exceeds Smillie-Mennillo”) are qualitative descriptions of $n=3$ observations, not statistical claims. The Methods text states this explicitly; the headline framing in Results uses qualitative language (“two of three atlas pairs show strong concordance at broad tier” rather than “we observe a significant pattern of cross-atlas variation”).

Donor-LOO uncertainty intervals on broad-tier headline metrics under de Lange for both methods. Liu sensitivity reported as point estimates only. Reported as “95% LOO jackknife range,” not “95% CI.” For Kong 2023 specifically (12 donors total, 6 UC + 6 healthy), enforce a minimum-after-LOO threshold of ≥ 5 cases per group (drop LOO iterations that violate this).

2.6 Cell-type label harmonization

One vocabulary, gut mucosa, two granularity tiers, both primary in locked v1:

- **Broad (~10-15 categories):** T cell, B cell, plasma cell, monocyte/macrophage, dendritic cell, mast cell, NK cell, ILC, enterocyte, goblet cell, stem/TA, fibroblast, endothelium, smooth muscle.
- **Fine (~30-50 categories):** subdivided by activation state, lineage subset, etc.

Why fine tier is core. UC pathology lives at fine granularity — BEST4+ enterocytes, inflammatory fibroblasts, IL-17+ T cells.

Cross-tissue extension architecture (v2 setup decision). v1’s vocabulary is gut mucosa. v2 extends to RA synovium, which has different cell types entirely (synovio-cytes, sublining/lining fibroblasts). The harmonization vocabulary needs three tiers in v2 — and v1 should label its immune cells in a way that makes the v2 cross-tissue tier possible:

- *v2 cross-tissue tier* (built later): shared immune lineages — T cell, B cell, plasma cell, monocyte/macrophage, DC, mast, NK. Usable across gut and synovium.
- *Within-tissue broad tier* (v1’s primary): tissue-specific broad categories.
- *Within-tissue fine tier* (v1’s primary): fine cell states.

v1 architectural decision: when manually mapping immune cells in M2, prefer **Cell Ontology terms whose parent terms align with synovium-applicable categories**. Concrete rule: if the gut crosswalk maps “Inflammatory monocyte” to a CL term like CL:0002054 (“nonclassical monocyte”), check that the parent CL chain leads through CL:0000576 (“monocyte”) — a term that admits both gut and synovium daughters. If you’d be tempted to use a leaf-level CL term that’s too gut-specific, step up one level. Cost: ~1 day of CL ontology browsing during M2.

For v1 figures and analysis, immune cells continue to use the same labels as before — this decision affects only the *underlying CL parent assignment* in the crosswalk CSV, which is invisible in v1 outputs but available for v2 lookup.

Three strategies for harmonization, in order: scOntoMatch + Cell Ontology → manual CSV crosswalk → marker-gene confirmation. Plan ~5–7 days for both tiers. Pan-GI and HCA Gut (restricted to large intestine) join the crosswalk.

Sub-clustering depth at fine tier — roll up to lowest-resolution atlas. Atlases have different fine-tier sub-clustering depths: Smillie 2019 reports ~50 fine-tier states; Kong 2023 reports ~30; Mennillo may report ~25. When Smillie has CD4 Treg, CD4 Tcm, CD4 Tem, CD4 Th17 as separate fine labels but Kong has just “CD4 T” as one label, the fine-tier vocabulary uses **the depth of the lowest-resolution atlas in any pairwise comparison**. Smillie’s higher-resolution sub-states are aggregated up to “CD4 T” for cross-atlas concordance with Kong. The higher-resolution labels are reported in supplementary tables (Smillie-only fine-tier within-atlas analysis), not used in headline metrics. Locked in DECISIONS.md.

Marker-gene quality control at end of M2. After the crosswalk is built, validate that harmonization is biologically coherent. For each broad cell type in the harmonized vocabulary, define a panel of canonical markers (e.g., T cell: CD3D, CD3E, CD4, CD8A; B cell: CD19, MS4A1; plasma: MZB1, JCHAIN; enterocyte: VIL1, FABP1, KRT20; goblet: MUC2, TFF3; monocyte/macrophage: CD68, CD14, FCGR3A; mast: TPSB2, KIT; etc.). Compute mean expression of each cell type’s marker panel within each atlas’s harmonized cell type. **Flag any cell type where one atlas’s mean marker expression deviates by >2 SD from the other atlases’ mean** — this indicates the harmonized label is biologically inconsistent across atlases. Investigate flagged cells (likely a sub-population mis-mapping in the crosswalk) and re-map before locking the crosswalk. ~1 day at end of M2. Locked in DECISIONS.md.

Hard deadlines: broad-tier end of M2 (including marker-gene QC); fine-tier mid-M3.

2.7 Brown’s method for regime 2

Why Brown’s, not Fisher’s. Atlas-level p-values are not independent (shared MAGMA inputs, shared etiology, partially shared donors). Naive Fisher’s under-states effective evidence and gives anti-conservative combined p-values.

Implementation: analytical Kost-McDermott via EmpiricalBrownsMethod::kostsMethod(), with the cross-atlas correlation matrix estimated empirically from null draws. Run separately for each (method, tier) under de Lange — 4 regime-2 outputs.

Cross-atlas correlation matrix estimation. Kost-McDermott requires an estimate of the correlation matrix between the test statistics being combined. We estimate this matrix **empirically from the per-null-draw cell-type-level statistics** generated in M3 (scDRS Monte Carlo null draws; seismicGWAS gene-Z permutations — see §2.4.1 and §2.4.2 for the serialization format). For each cell type c and atlas pair (A, B), we have N null statistic pairs ($T_c^A_i, T_c^B_i$) for $i = 1..N$ ($N = 1000$). The Pearson correlation of these pairs is the (A, B) entry for cell type c . The Kost-McDermott combination is then performed per cell type using that cell-type-specific correlation matrix.

This is *the* methodological commitment that distinguishes “thoughtful Brown’s” from “ad hoc analytical procedure.” It is locked in DECISIONS.md at M0 and pre-specified in Methods.

Sanity check on the empirical correlation matrix. Off-diagonal entries should be positive (atlases share GWAS inputs and tissue biology). If any cell type has near-zero or negative cross-atlas null correlation, that's a debugging signal — most likely indicates harmonization mismatch (cells labeled “T cell” across atlases that are actually different populations) or insufficient sample size for the cell type. Investigate before proceeding to the combination step.

Edge case fallback for low-variance null statistics. For very small cell types, a single atlas's null statistic may have near-zero variance (all 1000 null draws produce nearly identical mean Z-scores), making the Pearson correlation undefined or unstable. **Fallback rule:** for any cell type where the null-statistic SD in any atlas falls below the 5th percentile of null-statistic SDs across all cell types in that (method, tier, GWAS) combination, replace the cell-type-specific correlation matrix with the **median cross-atlas correlation across well-behaved cell types** (those above the 5th percentile threshold) in that combination. The 5th-percentile cutoff is data-adaptive (calibrated per-comparison rather than hardcoded) and flags the bottom-tail cell types where the empirical correlation estimate is least reliable. Affected cell types flagged in supplementary tables. Locked in DECISIONS.md.

Heterogeneous per-atlas sample sizes. Smillie has 30 donors, Kong has 12, Mennillo (pre-treatment-only) ~10-15 donors. Per-atlas effective sample sizes differ by ~2-3x. Standard meta-analysis would inverse-variance-weight or sqrt(N)-weight (Stouffer's). Brown's-on-p-values via `kostsMethod()` does not apply per-study weights; it treats the inputs as equally informative. **Pre-commit:** use unweighted Brown's via `kostsMethod()` and document the equal-weighting as an approximation defensible because per-atlas N ratios are <3x. If reviewers push back, implement Stouffer-weighted Brown's in revision (~1 day of custom code). Locked in DECISIONS.md.

Stretch #1 (permutation-based Brown's combination) is a conditional fallback, not a mandatory sensitivity check. Activate if the empirical correlation matrix shows pathological entries (negative or near-zero off-diagonals where the shared MAGMA inputs and shared etiology guarantee positive). With the correlation matrix now estimated empirically rather than via theoretical assumption, the analytical Kost-McDermott approach is much more defensible than it was under the original under-specified plan, and the fallback is unlikely to be needed.

Cell types missing from some atlases (fine tier): - 3/3 atlases: standard Brown's combination using full 3×3 empirical correlation matrix. - 2/3: combine with `n_atlases = 2` flag using the relevant 2×2 correlation submatrix. - 1/3: do not combine; report regime-1 with `n_atlases = 1`. Exclude from regime-2 ranking.

2.8 Sanity scaffolding

Test-retest baseline. 3 seeds × 3 atlases × 2 methods = 18 runs (under de Lange only).

Gates: scDRS pass if $\rho \geq 0.9$; seismicGWAS pass if $\rho \geq 0.999$ (deterministic given fixed cell-type labels and expression matrix).

scDRS positive control: Tabula Muris × Yengo 2022 height GWAS. Confirms scDRS *can* detect signal (modulo mouse-human ortholog mapping caveat).

scDRS negative control: schizophrenia GWAS on Smillie at broad tier. Run scDRS with **Trubetskoy 2022 schizophrenia GWAS** (*Nature* 604:502–508) as the trait against Smillie. Schizophrenia is a clearly non-colon trait; we expect no colon cell type to achieve FDR<0.05 enrichment. If colon cell types do enrich for schizophrenia, the pipeline is producing spurious signal — debug before proceeding. ~1 additional scDRS run. Complements the height positive control by showing scDRS *doesn't manufacture* signal.

Caveat on schizophrenia negative control purity. Schizophrenia GWAS hits include MHC-region signal and complement-pathway genes, which can theoretically enrich for antigen-presenting / immune cell types in non-brain tissues. We mitigate this by (a) excluding MHC genes from all scDRS gene sets including the schizophrenia control (per §2.2 MHC policy) and (b) noting in Methods that “absence of enrichment” rather than “exact null” is the operational expectation. A perfectly clean negative control (e.g., hair color or LDL on non-hepatocyte cells) is deferred to revision if reviewers ask. Documented in DECISIONS.md.

MAGMA gene-property sanity track on Smillie at broad tier under de Lange. Watanabe-style regression. Validates upstream pipeline.

2.9 Donor-level leave-one-out uncertainty intervals

Donor-LOO drops each donor in turn, re-runs the method, recomputes cell-type p-values at both granularities, recomputes concordance metrics.

Donor counts: Smillie ~30, Kong-UC ~12, Mennillo ~20.

Scope: broad-tier headline metrics under de Lange for both methods. Liu = point estimates only. Report as “95% LOO jackknife range,” not “95% CI.”

Compute: 12–30 donors × 3 atlases × 2 methods = up to 180 LOO runs. ~90 hours.

2.10 Compute budget — locked v1 with stretches

Source	Locked v1	If stretches activate
scDRS regime-1 baseline (3 atlases × 2 GWAS)	6	—
scDRS test-retest (3 seeds × 3, de Lange only)	9	—
scDRS donor-LOO (12–30 donors × 3, de Lange only)	up to 90	—
scDRS HCA Gut (large-intestine subset, 2 GWAS)	2	—

Source	Locked v1	If stretches activate
scDRS Pan-GI (large-intestine subset, 2 GWAS)	2	—
scDRS Tabula Muris × height (positive control)	1	—
scDRS schizophrenia × Smillie (negative control)	1	—
scDRS Smillie × de Lange with MHC included (sensitivity)	1	—
scDRS Brown's permutation (conditional fallback if empirical Kost-McDermott pathological)	0	+300 (stretch #1)
scDRS regime 3 build	0	+20-30 (stretch #2)
scDRS LDL × TS hepatocytes	0	+1 (stretch #3)
Total scDRS runs	~112	~442
At 30-60 min/run	55-110 hours	220-440 hours
seismicGWAS regime-1 baseline (3 × 2 GWAS)	6	—
seismicGWAS test-retest (3 × 3, de Lange)	9	—
seismicGWAS donor-LOO (12-30 × 3, de Lange)	up to 90	—
seismicGWAS HCA Gut (2 GWAS)	2	—
seismicGWAS Pan-GI (2 GWAS)	2	—
seismicGWAS gene-Z permutation null draws (for empirical correlation matrix)	~5 hours equivalent	—
Total seismicGWAS runs	~109 + ~5 hours permutation	~109 + ~5 hours permutation
At 5-15 min/run	~14-32 hours	~14-32 hours
Combined locked v1 compute	~69-142 hours	~234-472 hours

Pan-GI is large (~1.1M cells); scDRS may need ~30 GB memory and ~1.5-2 hours per run. Account for higher-memory nodes in Hummingbird allocation.

Descope order: stretches → drop seismic donor-LOO → reduce scDRS donor-LOO. Do not drop scDRS broad-tier donor-LOO under de Lange.

2.11 Tooling stack

- **Python 3.10** (verify scDRS support before 3.11+)
- **scDRS** — `pip install scdrs==1.0.4` (verify in M1)
- **seismicGWAS** — `devtools::install_github("ylaboratory/seismicGWAS")`
- **MAGMA** — precompiled binary from CTGLab
- **plink 1.9**
- **scanpy** for AnnData
- **scOntoMatch** — `devtools::install_github("Papatheodorou-Group/scOntoMatch")`
- **EmpiricalBrownsMethod** R package (`kostsMethod()`)
- **anndata2ri** or **zellkonverter** — AnnData ↔ SingleCellExperiment
- **pandas, numpy, scipy, statsmodels, scikit-learn, matplotlib, seaborn**
- **R 4.3+** with **tidyverse, ggplot2, ggsci, ComplexHeatmap, ontologyIndex, SingleCellExperiment**

Reproducibility. `pyproject.toml`, `renv.lock`, `Dockerfile`.

Part 3: Datasets

3.1 GWAS

Locked v1 primary: de Lange et al. 2017 UC. GWAS Catalog GCST004131. ~12,366 cases / ~33,609 controls, EUR.

Locked v1 cross-GWAS sensitivity: Liu et al. 2023. *Nat Genet* 55:796–806. Multi-ancestry IBD meta-analysis, UC arm only. Verify per-SNP N column availability in M1; document fixed-N approximation in DECISIONS.md if absent.

Negative control: Trubetskoy 2022 schizophrenia GWAS. *Nature* 604:502–508. Run on Smillie only at broad tier under de Lange’s pipeline. Sanity check that no colon cell type achieves FDR<0.05 enrichment.

MHC region excluded from scDRS top-1000 gene set and seismicGWAS Z-score table, with sensitivity analysis (MHC-included) supplementary. **Autosomes only** (chr 1–22). See §2.2.

Do not use Sazonovs 2022 (CD-only rare-variant burden — different model).

3.2 scRNA-seq atlases

UC trio (cross-atlas axis):

- **Smillie et al. 2019.** *Cell* 178:714–730. 366,650 colon mucosa cells, 18 UC + 12 healthy. Single Cell Portal SCP259.
- **Kong et al. 2023 (UC subset).** *Immunity* 56:444–458. 6 healthy + 6 active UC. GEO GSE214695.
- **Mennillo et al. 2024.** *Nat Commun* 15:1493. Anti-integrin therapy time course; **use pre-treatment baseline samples only** (~10–15 donors after subsetting; verify in M1, drop to fallback if N<8). GEO accession to confirm in M1.

Atlas preprocessing policy: take published preprocessing as-is (Option B, §2.3); do not re-run uniform doublet removal / ambient correction / QC. Disclose preprocessing heterogeneity in Methods Limitations.

Pre-identified fallbacks: Garrido-Trigo 2023, Boland 2020, Devlin/Zhao 2023.

Broad-atlas comparators (two complementary roles):

- **HCA Gut Cell Atlas / Elmentaite et al. 2021.** *Nature* 597:250–255. ~140k healthy intestinal cells. **No donor overlap with UC trio.** cellxgene. Role: independent broad comparator.
- **Pan-GI / Oliver et al. 2024.** *Nature* 635:699–707. ~1.1M cells, 189 donors. **Substantial donor overlap with UC trio: integrated Smillie 2019, anchored Kong 2023.** cellxgene. Role: integration-pipeline-robustness comparator.

The two broad comparators play different epistemic roles. HCA Gut tests broad-atlas substitution; Pan-GI tests integration-pipeline robustness. Reviewers should not read Pan-GI as a second independent replication.

3.3 Reference data

- **1000 Genomes Phase 3 EUR** (MAGMA LD reference for both GWAS).
 - **Cell Ontology OWL file** (scOntoMatch + cross-tissue parent-term checks for v2 setup).
 - **HGNC gene-symbol release** pinned (e.g., HGNC 2024-Q1).
-

Part 4: Month-by-Month Plan

12 months, two authors, gates at M3, M5, M7.

M0 (pre-start): planning lock-in

- Two-author division of labor in README. Authorship plan agreed with PI mentor.
- DECISIONS.md committed and dated.
- **PI commitment to v2 trajectory explicit.** If yes, v1 includes the three v2-setup decisions (CL-ontology-aware harmonization in M2; specificity score serialization in M3; generalized donor-attribution metadata in M1). If no, drop the three setup decisions; v1 stands alone.
- GitHub repo, environment files, directory structure.
- Hummingbird allocation verified against ~110 scDRS + ~109 seismic = ~64–137 node-hours. Pan-GI may push some runs to higher-memory nodes.
- Reading list M1–M2 papers identified.
- PI mentor sign-off on locked-core scope and stretch ladder.

M1: environment, GWAS, atlases

Goal: working scDRS + seismicGWAS pipelines on tutorial data; clean MAGMA .gs files for de Lange and Liu; 5 atlases standardized; donor-attribution infrastructure in place.

- Both authors set up dev environments. Run scDRS + seismicGWAS tutorials.
- MAGMA pipeline ×2: download de Lange UC and Liu 2023 UC. QC each (compute λ_{GC} and document). Run MAGMA SNP-to-gene + gene-based test against 1000G EUR, **autosomes only (chr 1-22)**. Convert top 1,000 to scDRS .gs for each, **excluding MHC genes (chr 6: 28,477,797-33,448,354 in GRCh37)**. Sanity-check top genes for both. Document the EUR-LD-reference-for-multi-ancestry-Liu choice in DECISIONS.md.
- **Download Trubetskoy 2022 schizophrenia GWAS** (negative control). MAGMA pipeline applied identically (autosomes, MHC-excluded, 1000G EUR LD reference acknowledged as approximation for psychiatric trait but acceptable for a negative-control baseline).
- Atlas downloads: Smillie (SCP259), Kong (GSE214695), Mennillo (verify accession, **subset to pre-treatment baseline samples only — verify N≥8 donors after subsetting**), HCA Gut (cellxgene, large-intestine subset), Pan-GI (cellxgene, large-intestine subset). Convert each to standardized .h5ad with HGNC-pinned symbols.
- **Per-atlas covariate file generation (<atlas>_covariates.tsv)**. For each atlas, generate the scDRS covariate file with columns: cell_id, log_n_genes, log_n_counts, donor (one-hot), sample (one-hot if multiple samples per donor), **sex (one-hot, if donor-level sex metadata exists; document omission in DECISIONS.md if missing for any atlas)**. Apply consistently across all 5 atlases. Locked covariate set documented in DECISIONS.md.
- **Generalized donor-attribution metadata (v2 setup decision)**. For every atlas, save donor_metadata.csv with columns donor_id, originating_study, tissue, disease_status, assay_protocol. Pan-GI is the immediate consumer (donor-overlap exclusion at M5); the same schema scales to AMP-RA Phase 2 and any future integrated atlas in v2. For non-integrated atlases, originating_study is just the atlas name.
- **Inflamed/uninflamed sample composition documented per atlas** in donor_metadata.csv extension (column tissue_state ∈ {inflamed, uninflamed, unspecified}). Methods Table 1 derives from this.

Spot-checks:

- GWAS Catalog GCST004131 resolves correctly.
- Liu 2023 UC arm cleanly separable from CD and IBD-combined arms.
- **Liu 2023 per-SNP N column present** (or document fixed-N approximation in DECISIONS.md).
- $\lambda_{GC} \leq 1.1$ for both de Lange and Liu (or flag for revision response if higher).
- **Trubetskoy 2022 schizophrenia GWAS download successful** and MAGMA pipeline runs cleanly under autosomes-only, MHC-excluded settings.
- scdrs==1.0.4 still installable; if newer versions, evaluate.
- seismicGWAS install completes; tutorial runs end-to-end.

- Mennillo 2024 GEO accession confirmed; **pre-treatment subsetting yields ≥ 8 donors**. If < 8 , swap to fallback (Garrido-Trigo 2023).
- `EmpiricalBrownsMethod::kotsMethod()` API exists.
- Pan-GI donor metadata sufficient to identify Smillie 2019 and Kong 2023 contributors. **If insufficient, drop Pan-GI to stretch.**
- **HCA Gut donor metadata cross-checked against Smillie 2019, Kong 2023, Mennillo 2024 donor IDs — verify zero overlap.** HCA Gut Cell Atlas (Elmentaite 2021) integrates multiple gut atlases; verify its donor set excludes our UC trio. If HCA Gut overlaps with our trio, it becomes a second integration-pipeline-robustness comparator (not an independent comparator), and the §2.4.3 framing changes accordingly.
- **Pan-GI and HCA Gut raw count layers accessible.** Verify `adata.layers['counts']` (or equivalent) contains raw counts for each. If only normalized/integrated counts are accessible, drop to stretch — scDRS requires raw counts.
- **seismicGWAS confounder defaults inspected.** Open the seismicGWAS package source; confirm `get_ct_trait_associations()` defaults include gene length, gene-gene LD score, and transcript count. If defaults differ from this set, override and apply explicitly. Document in DECISIONS.md.
- **MHC gene exclusion verified** in both .gs files and Z-score tables for de Lange + Liu + Trubetskoy.
- **Cell Ontology release pinned in DECISIONS.md** (record the OWL file release date downloaded for M2 work).

Deliverable: scDRS + seismicGWAS on tutorial; three MAGMA .gs files + Z-score tables (de Lange, Liu, schizophrenia negative control), MHC-excluded; 5 atlases standardized with Mennillo subsetting to pre-treatment baseline; per-atlas donor metadata in standardized format including inflamed/uninflamed status; Hummingbird allocation verified; DECISIONS.md committed.

M2: cell-type harmonization

Goal: broad-tier crosswalk for gut mucosa vocabulary jointly validated across 5 atlases. Fine-tier crosswalk substantially underway. **CL-ontology-aware immune-cell mapping completed.**

1. **Broad-level mapping (10-15 cell types).** `scOntoMatch` + Cell Ontology where possible; manual mapping otherwise. Verify with marker genes. Crosswalk has columns `atlas`, `original_label`, `broad`, `fine`, `cl_term`, `cl_parent_chain` for all 5 atlases.
2. **Fine-level mapping (30-50 cell states), targeted for completion mid-M3.** Sub-clustering depth rolls up to lowest-resolution atlas in any pairwise comparison (per §2.6). Higher-resolution sub-states retained in supplementary tables, not used in headline metrics.
3. **Joint validation.** Both authors review each other's mappings.
4. **CL-ontology cross-tissue check (v2 setup decision, ~1 day).** For every immune cell type in the crosswalk, verify the assigned CL term has a parent that admits both gut and synovium daughters. Walk the CL parent chain until you hit a tissue-agnostic immune lineage term. Document the chain in `cl_parent_chain` column. If any immune cell type's CL assignment is gut-specific (e.g., a leaf

term that has no synovium counterpart in CL), step up one level. Invisible in v1 outputs — lives in the crosswalk CSV — but available for v2 lookup.

5. **Marker-gene quality control (~1 day, end of M2).** For each broad cell type in the harmonized vocabulary, define a panel of canonical markers (T cell: CD3D, CD3E, CD4, CD8A; B cell: CD19, MS4A1; plasma: MZB1, JCHAIN; enterocyte: VIL1, FABP1, KRT20; goblet: MUC2, TFF3; monocyte/macrophage: CD68, CD14, FCGR3A; mast: TPSB2, KIT; NK: NCAM1, NKG7; ILC: KIT, IL7R without CD3; stem/TA: LGR5, OLFM4, MKI67; fibroblast: COL1A1, COL3A1; endothelium: VWF, PECAM1; smooth muscle: ACTA2, MYH11). Compute mean expression per atlas per cell type. Flag cell types where one atlas deviates by >2 SD from others on canonical-marker expression. Investigate flagged cells; re-map crosswalk where biological mismatch is identified. Lock the crosswalk only after marker-gene QC passes.

Risk mitigation: weekly check-ins; identify by mid-M2 whether either author needs help. If broad-level isn't done by end of M2, descope at M3.

Deliverable: broad-tier crosswalk CSV with CL-term and parent-chain columns, jointly validated across 5 atlases, **with marker-gene QC passed**; fine-tier crosswalk substantially underway with sub-clustering depth rolled up to lowest-resolution atlas.

M3: regime 1 (both methods, both GWAS) + sanity scaffolding + fine-tier completion [Gate 1]

Goal: complete regime-1 scDRS *and* seismicGWAS analyses across UC trio at both granularity tiers, both GWAS; positive control passes; MAGMA gene-property sanity passes; per-(atlas, method) test-retest passes; **specificity scores serialized in comparison-ready format.**

Track A — Regime 1 scDRS:

- scDRS on 3 UC atlases × 2 GWAS = 6 baseline compute-score runs. Group analysis at both tiers (post-processing).
- Test-retest under de Lange: 3 seeds × 3 atlases = 9 runs.
- **Aggregate per-null-draw cell-type Z-scores per (atlas, GWAS, tier).** scDRS Monte Carlo internally generates ~1000 null gene-set draws; aggregate per-cell null Z-scores to cell-type level (mean per-cell Z within cell type, per null draw). Save as feather files indexed by (null_draw_idx, cell_type). These are the inputs to Brown's empirical correlation matrix in M5. Cost: post-processing on existing scDRS outputs, ~30 min total.

Track B — Regime 1 seismicGWAS + sanity scaffolding:

- seismicGWAS specificity scores per atlas at both tiers (GWAS-independent): 6 specificity computations.
- **Specificity score serialization (v2 setup decision).** Save specificity as long-format .rds (or feather) per (atlas, granularity, cell_type, gene). One file per atlas. Stored alongside regression results, not embedded in them. Cost: ~half a day of pipeline work.
- seismicGWAS regressions: 3 atlases × 2 GWAS × 2 tiers = 12 regressions (counted as 6 baseline runs for budget). **Confounders applied: gene length,**

gene-gene LD score, transcript count (verified in M1).

- **Generate gene-Z permutation null draws per (atlas, GWAS, tier).** $M = 1000$ permutations of the gene-Z-score vector; re-run `get_ct_trait_associations()` under each; save per-permutation cell-type-level test statistics to feather. Cost: ~30 min per (atlas, GWAS, tier); ~5 hours total. These are the inputs to Brown's empirical correlation matrix for the seismicGWAS regime-2 combination in M5.
- Test-retest under de Lange: 3 seeds $\times$ 3 atlases = 9 runs. **Test-retest gate tightened to $\rho \geq 0.999$** for seismicGWAS — the method is deterministic given fixed cell-type labels and a fixed expression matrix, so test-retest should be ≈ 1.0 ; deviation indicates a bug.
- MAGMA gene-property sanity track on Smillie at broad tier under de Lange.
- scDRS positive control on Tabula Muris $\times$ Yengo 2022 height GWAS.
- **scDRS negative control: schizophrenia GWAS on Smillie at broad tier.** 1 additional compute-score run + group analysis. Confirm no colon cell type achieves $FDR < 0.05$ enrichment. If any do, debug pipeline before proceeding.
- Complete fine-tier harmonization (rolling over from M2).

Test-retest gates:

- scDRS: pass ≥ 0.9 , amber 0.7-0.9, fail < 0.7 .
- seismicGWAS: pass ≥ 0.999 , amber 0.99-0.999, fail < 0.99 (deterministic method; deviation indicates bug).

Gate 1: "Is regime 1 working for both methods, both GWAS, both tiers?"

- Test-retest passed for both methods?
- MAGMA gene-property on Smillie produced sensible cell-type ordering?
- Tabula Muris $\times$ height GWAS positive control passed?
- Fine-tier harmonization complete?
- Liu 2023 regime-1 produced non-degenerate rankings (not all clustered at $p=1.0$ or $p=0.0$)?
- **Specificity scores saved in comparison-ready format?**

If yes to all: proceed to M4. If any fails: stop and debug.

Deliverable: regime-1 cell-type rankings for 3 UC atlases $\times$ 2 methods $\times$ 2 granularities $\times$ 2 GWAS = 24 ranking sets; test-retest baselines; positive control passed; MAGMA gene-property sanity passed; fine-tier harmonization complete; specificity scores serialized; DECISIONS.md updated.

M4: concordance analysis (5 axes) + introduction draft

Goal: all five concordance axes computed for regime 1 with bootstrap 95% CIs; BH-FDR correction applied across the secondary comparison battery; donor-LOO uncertainty intervals on broad-tier de Lange headline metrics; introduction drafted.

- Cross-atlas concordance: 24 ranking comparisons. **Spearman ρ on Z-scores (scDRS) and regression coefficients (seismicGWAS), on shared cell-type intersection per pair.**
- Cross-method concordance: 12 within-atlas scDRS-vs-seismic comparisons.
- Cross-GWAS concordance: 12 within-atlas de Lange-vs-Liu comparisons.

- **Bootstrap 95% CIs on every reported Spearman ρ — 1000 iterations, re-sampling over cell types within shared intersection.** Cheap (~seconds per comparison).
- **Benjamini-Hochberg FDR < 0.05 across the full secondary comparison battery (~60 comparisons).** Single primary analysis (3×3 atlas-pair × broad-tier × scDRS × de Lange) reported uncorrected. Uncorrected p-values in supplementary tables.
- Donor-LOO intervals on broad-tier headline metrics under de Lange for both methods (~180 LOO runs, with Kong minimum-cases enforcement from §2.5).
- **Headline figures:**
 - 3×3 atlas-pair Spearman ρ heatmap with bootstrap CIs, faceted by method × granularity × GWAS.
 - Cell-type forest plot for broad tier (de Lange) with donor-LOO ranges.
 - Cell-type forest plot for fine tier (de Lange) with donor-LOO ranges.
 - Cross-method concordance comparison panel with bootstrap CIs.
 - Cross-GWAS concordance comparison panel with bootstrap CIs (de Lange vs Liu within atlas).

Joint task: start drafting introduction.

Deliverable: 5 concordance axes computed for regime 1 with bootstrap CIs, donor-LOO ranges, and BH-FDR-corrected significance for secondary claims; 5 headline figures publication-quality; introduction draft underway.

M5: regime 2 + broad-atlas comparators (HCA Gut + Pan-GI) + methods draft [Gate 2]

Goal: combined cell-type p-values via Brown’s method (analytical, de Lange only); both broad-atlas comparators completed including Pan-GI donor-overlap analysis; methods drafted.

Track A — Regime 2:

- **Compute empirical cross-atlas correlation matrices from M3 null-draw aggregations.** For each (method, tier) combination under de Lange, build the per-cell-type empirical correlation matrix from the saved null statistics: scDRS uses the Monte Carlo null aggregations, seismicGWAS uses the gene-Z permutation aggregations (both serialized in M3). Sanity check: off-diagonal entries should be positive; investigate any cell type with near-zero or negative cross-atlas null correlation before proceeding.
- `EmpiricalBrownsMethod::kotsMethod()` for each (method, tier) under de Lange, **using the empirical correlation matrix.** 4 regime-2 outputs.
- If empirical correlation matrix is pathological (negative or near-zero off-diagonals), activate stretch #1 (permutation Brown’s) as fallback.
- Regime-1-vs-regime-2 comparison figures with bootstrap CIs.

Track B — Broad-atlas comparators:

- scDRS + seismicGWAS on HCA Gut (large-intestine subset), both GWAS. 4 runs.
- scDRS + seismicGWAS on Pan-GI (large-intestine subset), both GWAS. 4 runs.
- HCA Gut: independent broad-atlas concordance with UC trio.

- **Pan-GI dual analysis:**
 - With donor overlap retained: report concordance with each UC atlas; this includes shared donors with Smillie + Kong (sanity check).
 - With donor-overlap exclusion: subset Pan-GI to donors not from Smillie 2019 or Kong 2023 (use the donor-attribution metadata from M1). Re-compute concordance. **This is the headline Pan-GI panel.**
- Build sensitivity figure showing Pan-GI concordance with vs without donor-overlap exclusion.
- Cross-atlas donor audit across all 5 atlases.

This month is heavy on implementation but light on scDRS compute. Use slack to draft methods.

Gate 2: “Is the locked core clean?”

- All locked-core figures publication-quality?
- Spearman ρ , Jaccard, κ tables exist with donor-LOO ranges?
- Regime-1-vs-regime-2 comparisons have a clear story?
- HCA Gut independent-broad-atlas comparison clean?
- Pan-GI donor-overlap analysis clean (with and without exclusion)?
- Cross-method comparison panel done?
- Cross-GWAS comparison panel done?

If yes: proceed to M6 stretch evaluation. If no: M6 finishes locked core; defer all stretches.

Deliverable: regime-2 combined rankings (4 method $\times$ tier under de Lange); regime-1-vs-regime-2 figures; HCA Gut + Pan-GI broad-comparator panels; cross-atlas donor audit; cross-method + cross-GWAS comparisons; methods section drafted; introduction draft complete.

M6: write-up first draft + stretch decision

Goal: locked-core figures and tables in publication-quality form; first complete manuscript draft; PI feedback round 1; stretch ladder evaluated.

- Complete results section (~8–10 figures).
- Discussion section with five angles: cross-atlas at broad and fine; cross-method; cross-GWAS; regime-1-vs-regime-2; HCA Gut vs Pan-GI.
- **Headline-claim lock-in.** Pick the one organizing claim of the paper (most likely: “fine-tier UC cell-type prioritization findings are not robust across atlases; broad-tier and meta-analyzed findings are”). Restructure paper around it. Other findings become supporting evidence. Document in DECISIONS.md.
- Send first complete draft to PI mentor.
- **Stretch evaluation.** Stretch ladder is 3 items.
 - *Locked core clean and 4+ months remaining:* green-light stretch #3 (LDL $\times$ TS hepatocytes); consider stretch #2 (regime 3) cautiously — pulling regime 3 into v1 weakens v2 differentiation.
 - *Locked core clean and 2–3 months remaining:* green-light stretch #3 only.
 - *Locked core needs polish:* defer all stretches.
 - Stretch #1 only if M5 found Kost-McDermott pathology.

- Stretch #2 generally deferred to v2.

Deliverable: first complete manuscript draft (locked core); PI feedback received; headline-claim lock-in committed; stretch ladder decision committed.

M7: top stretches if green-lit [Gate 3]

If stretch #1 (Brown's permutation): ~300 scDRS runs; permutation-based covariance; compare to analytical Kost-McDermott.

If stretch #3 (LDL × TS hepatocytes): ~1 day. scDRS on Tabula Sapiens hepatocytes with LDL GWAS. Confirms human-only code path.

If stretch #2 (regime 3): scANVI integration of UC trio. Material time investment. Generally deferred to v2.

Gate 3: "Is writing on track?"

- Manuscript draft complete?
- Figures publication-quality?
- PI mentor reviewed at least one full draft?

If yes: M8 for additional stretches and polish. If no: M8 for completing writing.

Deliverable: any green-lit stretches integrated; full draft complete.

M8: lower-priority stretches + polish

- Remaining green-lit stretches.
- Polish manuscript; second-round PI review.

Deliverable: stretches integrated; manuscript polished; supplementary materials drafted.

M9: pre-submission peer review + revision

- One statistical genetics reviewer + one single-cell reviewer (informal).
- Revise. Pay particular attention to whether reviewers correctly distinguish HCA Gut (independent) from Pan-GI (re-integration). If ambiguity, sharpen Methods.
- Update DECISIONS.md.

Deliverable: revised manuscript; supplementary materials complete.

M10: journal selection + format

- *Bioinformatics Advances* or *NAR GBI*.
- Format to journal specs.
- Tag GitHub repo with Zenodo DOI. Verify Docker reproducibility.

Deliverable: submission-ready manuscript and reproducible repo.

M11: submission

- bioRxiv (DOI within 1-2 days).
- Journal submission.

Deliverable: preprint live; journal submission in flight.

M12: respond to journal feedback

Deliverable: first journal-review-round revisions in flight.

Three explicit fork points for descope decisions

- **M3 fork.** If broad-tier harmonization isn't done by end of M2, drop fine-tier from primary status. If both methods aren't producing test-retest-passing results, drop seismicGWAS. If Liu 2023 regime-1 produces degenerate rankings, drop Liu from core. If specificity score serialization isn't working, drop the v2-setup decision.
- **M5 fork.** If Pan-GI donor-overlap analysis is unclear or compute-intensive, drop Pan-GI to a stretch. If regime 2 isn't clean, defer cleanup to M6.
- **M7 fork.** Drop all stretches; M7-M11 is pure writing.

Do NOT drop: test-retest baselines for both methods; scDRS positive control on Tabula Muris; donor-LOO on broad-tier scDRS headline under de Lange; DECISIONS.md updates.

The three v2-setup decisions are *all droppable* under schedule pressure without compromising v1. Drop priority: CL parent chain first (least painful retrofit) → generalized donor-attribution second (Pan-GI metadata still required regardless) → specificity score serialization last (most painful retrofit).

Minimum viable paper if everything decays: scDRS only, 3 UC atlases, broad granularity, regime 1 + regime 2 (analytical), MAGMA gene-property sanity, test-retest baseline, height-GWAS positive control. Still publishable in *Bioinformatics Advances*.

Repo directory structure

```
uc-cross-atlas-v1/
├── DECISIONS.md
├── README.md
├── data/
│   ├── gwas/                # de Lange + Liu 2023
│   ├── atlases/             # Smillie, Kong-UC, Mennillo, HCA Gut, Pan-GI
│   │   └── donor_metadata/  # per-atlas donor_metadata.csv (v2 setup pattern)
│   │       └── reference/   # 1000G EUR, CL ontology, HGNC pin
├── code/
│   ├── 01_magma/            # x2 GWAS
│   └── 02_atlas_prep/       # download, harmonize, HGNC remap, donor-
attribution
```

```

|   |— 03_scdrs/           # regime 1 (×2 GWAS, ×2 tiers)
|   |— 04_seismic/        # regime 1 (×2 GWAS, ×2 tiers)
|   |   |— specificity_long/ # serialized specificity for v2 reuse
|   |— 05_magma_geneprop/  # Watanabe-style sanity track
|   |— 06_concordance/     # 5 axes
|   |— 07_regime2_meta/    # Brown's method (analytical, de Lange only)
|   |— 08_cross_method/    # scDRS vs seismicGWAS within-atlas
|   |— 09_cross_gwas/      # de Lange vs Liu within-atlas
|   |— 10_broad_atlas_hca/ # HCA Gut comparator (independent)
|   |— 11_broad_atlas_pangi/ # Pan-GI comparator (with/without donor-
overlap)
|   |— 12_positive_control/ # Tabula Muris × height
|   |— 13_figures/
|— results/
|— notebooks/
|— manuscript/

```

Part 5: Risk Register

Risk	Probability	Impact	Mitigation
M2 broad-tier harmonization runs over M2 CL-ontology cross-tissue browsing eats more than 1 day	Medium	Medium	Weekly check-ins; mid-M2 deadline.
M3 fine-tier harmonization runs over M3 specificity serialization breaks pipeline	Low	Low	Time-box to 1 day. If running over, take crosswalk as-is; v2 retrofits later. v1 figures unaffected.
Compute exceeds Hummingbird allocation	Medium	Medium	Slip into M4 acceptable.
	Low	Low	Fallback: save specificity inside regression-result objects (default). v2 retrofits with one re-run per atlas later.
	Low-Medium	Low	Locked core ~64-137 node-hours. Pan-GI memory may need bigger nodes.

Risk	Probability	Impact	Mitigation
Liu 2023 GWAS download or QC issues	Medium	Medium	Drop Liu to stretch; ship de Lange-only locked core.
Liu 2023 EUR-LD criticized in review	Medium	Low	Document in Methods. Alternative is per-ancestry MAGMA + meta-combine, research project of its own.
Pan-GI donor metadata insufficient	Medium	High	Verify in M1. If too coarse, drop Pan-GI to stretch.
Pan-GI compute (1.1M cells) overruns memory	Medium	Medium	Test with realistic memory request in M5. May require subsampling or higher-memory node.
Reviewers conflate HCA Gut and Pan-GI roles	Medium	Medium	Sharpen Methods; pre-submission peer review at M9 tests for this.
PI not committed to v2 trajectory	Unknown	Medium	If unclear at M0, drop v2-setup decisions. v1 unaffected.
v2-setup decisions drift to v1 work without paying off	Low	Low	Time-box each: ~1 day for CL browsing, ~half-day for specificity, near-zero for donor-attribution beyond Pan-GI. If any consume more, drop.
seismicGWAS install / dependency hell	Medium	Low	Half-day in M1. Docker.
seismicGWAS produces nonsensical rankings on real data	Low	Medium	Cross-check against scDRS at broad tier on Smillie.

Risk	Probability	Impact	Mitigation
Two-author project decays into one-author	Medium	High	Document division in README; monthly PI check-ins.
Scope creep — year of analysis, three weeks of writing	High	High	Three gating decisions (M3, M5, M7). Drafting starts M4 (intro), M5 (methods), M6 (full draft).
One UC atlas inaccessible	Medium	Medium	Pre-identified fallbacks.
MAGMA gene-property fails on Smillie	Low	High	Gene-mapping broken — fix before proceeding.
scDRS positive control fails on TM	Medium	Medium	Mouse-human ortholog mapping path. Debug.
Kost-McDermott covariance pathological	Medium	Medium	Stretch #1 (permutation).
Cohen's κ marginal saturation at broad tier (modal)	High	Low	Pre-committed FDR<0.01 headline contingency.
Cross-GWAS concordance trivially high	Medium	Low	Likely outcome. Frame as confirmation, not headline.
Pan-GI with-donor-overlap concordance trivially high	High	Low	Itself the finding. Donor-overlap-excluded analysis is the headline.
Pan-GI donor-overlap-excluded subset too small	Medium	Medium	Pan-GI has 189 donors; ~50-80 from Smillie + Kong. Excluded subset ~100+ donors. Verify in M5.
Six months in, you discover a scoop	Low	High	Re-research literature at M4 and M7.
Author divergence on harmonization	Medium	High	Weekly sync meetings during M2-M3.

Risk	Probability	Impact	Mitigation
Manuscript writing slips past M11	Medium	High	Intro at M4; methods at M5; full draft at M6.

Part 6: Reading List

Must-read in M1:

- Watanabe K et al. 2019. *Nat Commun* 10:3222.
- **Skene NG et al. 2018. *Nat Genet* 50:825-833. [EWCE / specificity-based cell-type prioritization — methodological precedent for both seismicGWAS and our cross-atlas comparison framing.]**
- **Bryois J et al. 2020. *Nat Genet* 52:482-493. [Cross-atlas cell-type prioritization for psychiatric traits across human + mouse brain atlases — the foundational cross-atlas methodology paper. Cite as precedent in Introduction.]**
- Zhang MJ et al. 2022. *Nat Genet* 54:1572-1580. [scDRS paper.]
- Lai Q et al. 2025. *Nat Commun* 16:8744. [seismicGWAS paper.]
- Lakkis et al. 2024. *Frontiers in Immunology* 15:1399719.
- Smillie CS et al. 2019. *Cell* 178:714-730.
- de Lange et al. 2017. *Nat Genet* 49:256-261.
- Liu Z et al. 2023. *Nat Genet* 55:796-806.

Read in M2-M3:

- de Leeuw CA et al. 2015. *PLOS Comp Biol* 11:e1004219.
- Brown MB. 1975. *Biometrics* 31:987-992.
- Kost JT, McDermott MP. 2002. *Stat Probab Lett* 60:183-190.
- Yengo L et al. 2022. *Nature* 610:704-712. [Height GWAS — positive control on Tabula Muris.]
- **Trubetskoy V et al. 2022. *Nature* 604:502-508. [Schizophrenia GWAS — negative control on Smillie.]**
- Oliver AJ et al. 2024. *Nature* 635:699-707. [Pan-GI; pay attention to donor attribution.]
- **Diehl AD et al. 2016. “The Cell Ontology 2016: enhanced content, modularization, and ontology interoperability.” *J Biomed Semantics* 7:44. [For CL parent-chain navigation if doing the v2-setup CL browsing.]**

Read by M4:

- Li A et al. 2025. medRxiv 2025.05.24.25328275.
- Crow M et al. 2018. *Nat Commun* 9:884.
- Kong L et al. 2023. *Immunity* 56:444-458.
- Mennillo E et al. 2024. *Nat Commun* 15:1493.
- Elmentaite R et al. 2021. *Nature* 597:250-255.

Pre-v2 reading (deferred to v1 → v2 transition, M9-M12):

- Zhang F et al. 2019. *Nat Immunol* 20:928–942. [AMP-RA Phase 1, future v2 atlas.]
 - Zhang F et al. 2023. *Nature* 623:616–624. [AMP-RA Phase 2, future v2 atlas.]
 - Martin JC et al. 2019. *Cell* 178:1493–1508. [CD scRNA-seq, future v2 atlas.]
 - Okada Y et al. 2014. *Nature* 506:376–381. [Canonical RA GWAS.]
 - Ishigaki K et al. 2022. *Nat Genet* 54:1640–1651. [Multi-ancestry RA GWAS.]
-

Part 7: How to Know You’re On Track

- **End of M0:** DECISIONS.md committed; division of labor agreed; PI sign-off on v1 *and on v2 trajectory if applicable*; Hummingbird allocation verified.
- **End of M1:** scDRS + seismicGWAS tutorials reproduced; two MAGMA .gs files; 5 atlases standardized; **per-atlas donor metadata in standardized format**; spot-checks complete.
- **End of M2:** Broad-tier harmonization complete and jointly validated across 5 atlases; **CL parent-chain column populated for immune cell types (v2-setup)**; fine-tier crosswalk substantially underway. *Hard deadline.*
- **End of M3 (Gate 1):** Regime 1 results across 24 ranking sets; test-retest passed; positive control passed; MAGMA gene-property sanity passed; fine-tier harmonization complete; **specificity scores serialized in long-format files (v2-setup)**.
- **End of M4:** 5 concordance axes for regime 1 with donor-LOO ranges; 5 headline figures publication-quality; introduction draft underway.
- **End of M5 (Gate 2):** Regime 2 (Brown’s analytical) complete; regime-1-vs-regime-2 figures; HCA Gut + Pan-GI panels including Pan-GI donor-overlap sensitivity; cross-atlas donor audit; methods section drafted.
- **End of M6:** First complete manuscript draft; PI feedback round 1; **headline-claim lock-in committed in DECISIONS.md**; stretch ladder decision committed.
- **End of M7 (Gate 3):** Top stretches integrated if green-lit; full draft updated; PI feedback round 2.
- **End of M8:** Stretches integrated if green-lit; manuscript polished; supplementary materials complete.
- **End of M9:** Pre-submission peer review feedback incorporated.
- **End of M10:** Manuscript formatted; reproducible repo tagged with Zenodo DOI.
- **End of M11:** bioRxiv preprint live; journal submission in flight.
- **End of M12:** First-round journal revisions in flight.

If lagging more than two weeks, descope per M3 / M5 / M7 forks.

DECISIONS.md template (commit at M0)

```
# DECISIONS.md
```

Append-only log of analytical commitments. Never edit existing entries – only append, w

2026-XX-XX (M0): Locked v1 scope

****Project:**** Cross-atlas reproducibility of GWAS-driven cell-type prioritization in UC,

****Disease:**** Ulcerative colitis only. Cross-disease + cross-tissue (UC + CD + RA) gener

****GWAS (locked core):****

****GWAS (locked core):****

- Primary: de Lange et al. 2017 UC, GWAS Catalog GCST004131.
- Cross-GWAS sensitivity: Liu et al. 2023, UC arm only. Per-SNP N column verified in M1
- ****Negative control:** Trubetskoy 2022 schizophrenia GWAS** (run on Smillie at broad tie
- LD reference for both UC GWAS: 1000G EUR (acknowledged in Methods as approximate for
- ****MHC region (chr 6: 28,477,797–33,448,354 GRCh37) excluded**** from scDRS top-1000 gen
- ****Autosomes only**** (chr 1–22). X chromosome excluded.
- **** λ_{GC} verified ≤ 1.1 **** for both GWAS in M1 (or flagged for revision response if high
- Regime 2 on de Lange only.

****Atlases (locked core):****

- UC trio: Smillie 2019 (SCP259), Kong 2023 UC (GSE214695), Mennillo 2024 (GEO TBD, **p
- Independent broad comparator: HCA Gut Cell Atlas / Elmentaite 2021.
- Integration-pipeline-robustness comparator: Pan-GI / Oliver 2024 (with/without donor-
- Fallbacks: Garrido-Trigo 2023, Boland 2020, Devlin/Zhao 2023.
- ****Atlas preprocessing:**** take published preprocessing as-is (Option B). Doublet/ambie
- ****Inflamed/uninflamed sample composition**** documented per atlas in donor metadata; re

****Methods:**** scDRS and seismicGWAS in locked core. scPagwas not in v1 (compute infeasib

****Regimes:**** Regime 1 (single-atlas) + Regime 2 (Brown's-method analytical Kost-McDermo

****Granularity:**** Broad and fine tiers BOTH primary.

****Cell-type harmonization:****

- Vocabulary: gut mucosa.
- Strategy: scOntoMatch + manual CSV crosswalk + marker-gene confirmation, applied acro
- ****Sub-clustering depth:** roll up to lowest-resolution atlas in any pairwise comparison
- ****Marker-gene QC required at end of M2.**** Mean expression of canonical markers per at
- ****Cell Ontology release pinned**** in DECISIONS.md (record OWL file release date).
- Hard deadlines: broad-tier end of M2 (including marker-gene QC); fine-tier mid-M3.

****scDRS configuration:****

- ****Covariates in `compute-score`:**** `log\_n\_genes`, `log\_n\_counts`, donor (one-hot), sa
- ****All-cells policy:**** **`compute-score`** and group analysis run on all cells in each atl
- ****Top-1000 MAGMA gene cutoff**** for **`gs`** file (scDRS default), MHC-excluded. Sensitiv
- ****Headline random seed:** `seed=42`** for Monte Carlo null sampling. Test-retest uses s

```

**Statistical metrics:**
- **Cross-atlas headline metric: Spearman  $\rho$  on cell-type-level Z-scores (scDRS) and reg
```

- Cohen's κ on FDR-significance with marginals; top-5/10 Jaccard at broad, top-5/10/20
- **Tied-rank handling for Spearman ρ : average-rank tie-breaking (scipy/R default).**
- **Concordance computed on shared cell-type intersection per pair, with minimum cell-c
- **"Detected" operationally defined:** harmonized label present in atlas crosswalk AND
- **Bootstrap 95% CIs on every reported Spearman ρ : 1000 iterations, resampling over ce
- Five concordance axes: cross-atlas, cross-method, cross-GWAS, regime-1-vs-2, broad-co
- **Headline 3x3 heatmap is summarizing three pairwise ρ values.** Pattern claims across
- Donor-L00 uncertainty intervals on broad-tier headline metrics under de Lange for bot

```

**Multiple-testing strategy:**
- Single primary analysis: 3x3 atlas-pair Spearman  $\rho$  at broad tier under de Lange via s
- **All other secondary comparisons (~60 across the four other axes): Benjamini-Hochber
```

```

**Cohen's  $\kappa$  marginal-imbalance contingency:** if  $\geq 80\%$  of cell types pass  $FDR < 0.05$  in bo
```

```

**Pan-GI donor-overlap policy:** report with retained overlap (sanity check) and with c
```

```

**Pan-GI and HCA Gut counts policy:** raw count layer (`adata.layers['counts']`) from ce
```

```

**Brown's method (regime 2) implementation:**
- **Cross-atlas correlation matrix estimated empirically** from per-null-draw cell-type
- Per-cell-type correlation matrix (Pearson correlation across  $N=1000$  null draws between
- **Edge case fallback:** for cell types where null-statistic SD in any atlas falls bel
- **Heterogeneous per-atlas sample sizes (Smillie 30 / Kong 12 / Mennillo ~10–15 donors
- **Stretch #1 (permutation Brown's) is a conditional fallback** activated only if the
```

```

**seismicGWAS regression confounders:** gene length, gene-gene LD score, transcript cou
```

```

**seismicGWAS test-retest gate:**  $\rho \geq 0.999$  (deterministic method given fixed cell-type
```

```

**Sanity scaffolding (locked core, do not drop):**
- Test-retest baseline (3 seeds  $\times$  3 atlases  $\times$  2 methods, de Lange only).
- scDRS **positive** control on Tabula Muris  $\times$  Yengo 2022 height GWAS.
- **scDRS negative control on Smillie  $\times$  Trubetskoy 2022 schizophrenia GWAS at broad tie
- MAGMA gene-property sanity track on Smillie at broad tier under de Lange.
```

```

**Compute budget:**
- Locked v1: ~112 scDRS + ~109 seismic = ~69–142 node-hours.
- With all stretches: ~442 scDRS + ~109 seismic = ~234–472 node-hours.
- Hummingbird allocation as of [date]: [X] node-hours.
- Pan-GI may require ~30 GB memory.
- Descope order: stretches  $\rightarrow$  seismic donor-L00  $\rightarrow$  reduce scDRS L00. Do not drop scDRS br
```

```

**Stretch ladder (priority order):**
1. Brown's-method permutation-based covariance (conditional fallback if empirical Kost-
```

2. scANVI integration as regime 3 (deferred to v2 by default).
3. LDL × TS hepatocytes complementary positive control.

****Gating decisions:****

- Gate 1 (M3): regime 1 working for both methods, both GWAS, both tiers.
- Gate 2 (M5): locked core clean.
- Gate 3 (M7): writing on track.

****Authorship:**** documented in README. First author = primary manuscript drafter; co-first

****Two-author division of labor:**** documented in README.

v2 trajectory architectural decisions (commit at M0 if PI is committed to v2)

****v2 scope:**** UC + CD + RA cross-tissue generalization with regime 3 (scANVI integration)

****v1 architectural decisions to enable v2 cheaply:****

1. ****CL-ontology-aware harmonization (M2, ~1 day).****

- Crosswalk CSV includes ``cl_term`` and ``cl_parent_chain`` columns for every immune cell
- Immune-cell CL assignments use parent terms that admit both gut and synovium daughter
- Walk parent chain until tissue-agnostic; document chain in CSV.
- Time-boxed to 1 day during M2 harmonization. If running over, take crosswalk as-is
- Does not affect v1 figures or analysis.

2. ****seismicGWAS specificity score serialization (M3, ~half day).****

- Save specificity scores per (atlas, granularity, cell_type, gene) as long-format ``specificity_score``
- Schema: ``atlas``, ``cell_type``, ``granularity``, ``gene``, ``specificity_score``.
- Stored alongside regression results, not embedded in them.
- Required because v2 cross-tissue claim compares UC-T-cell specificity to RA-T-cell
- Embedding inside regression-result objects forces full re-run later.

3. ****Generalized donor-attribution metadata (M1, near-zero cost beyond Pan-GI).****

- Per-atlas ``donor_metadata.csv`` with columns: ``donor_id``, ``originating_study``, ``tissue``
- Pan-GI is the immediate v1 consumer (donor-overlap exclusion at M5).
- Same schema scales to AMP-RA Phase 2 and any future integrated atlas in v2.

****These three decisions are droppable**** under v1 schedule pressure without compromising

****v2 decisions deferred to v2 planning**** (typically M9–M12 of v1 when bandwidth permits)

- Specific RA atlases (likely AMP-RA Phase 1 + Phase 2; verify access at v2 M0).
- Specific RA GWAS (Okada 2014 vs Ishigaki 2022 multi-ancestry).
- CD atlas additions (Kong 2023 CD arm; Martin 2019; possibly Elmentaite 2020 CD subset)
- Three-tier harmonization architecture full design (cross-tissue / within-tissue broad)
- Regime 3 implementation choices (scANVI parameter tuning, integration QC metric thresh

```
- Whether to add a third author for v2 (compute and bandwidth scale meaningfully).  
  
**PI commitment status as of M0:** [explicitly recorded – yes/no/conditional]. If yes,  
  
---  
  
## [Append future entries here, dated, with rationale]
```

Final Note

This v1 plan is now substantial: five concordance axes, two methods, two GWAS, two tiers, two broad-atlas comparators, full sanity scaffolding, plus three lightweight v2-setup decisions that pay forward to a UC + CD + RA cross-tissue generalization paper. The headline-claim lock-in at M6 is the discipline that turns a comprehensive measurement set into a focused paper. Without it, the paper risks reading as “we measured all the things” rather than “we found that ___.”

The three v2-setup decisions are deliberately cheap. Together they cost ~1.5 days of v1 work plus a small architectural discipline in how outputs are saved. They unlock 18–24 months of v2 work without retrofit pain. *If* the PI is committed to v2. If not, drop them.

Mitigations: three gating decisions (M3, M5, M7), DECISIONS.md committed at M0, weekly sync meetings during M2–M3 harmonization, the discipline that stretches don’t ship as fallbacks for a slipping locked core, and the headline-claim lock-in at M6.

If you find yourself at M7 with no draft introduction, you are behind even if the analysis is going well. The paper is the deliverable.

This v1 ships in 2026 → v2 (UC + CD + RA cross-tissue with regime 3) ships in 2027–2028. Two papers on a single coherent research line, where v1 sets up v2 architecturally without paying v2 costs prematurely. Use the year you have.